# Predictors of COVID-19 Protective Behavior Before and After Mask Mandates: A Cross-National Analysis

Zixuan Zhao A1948883

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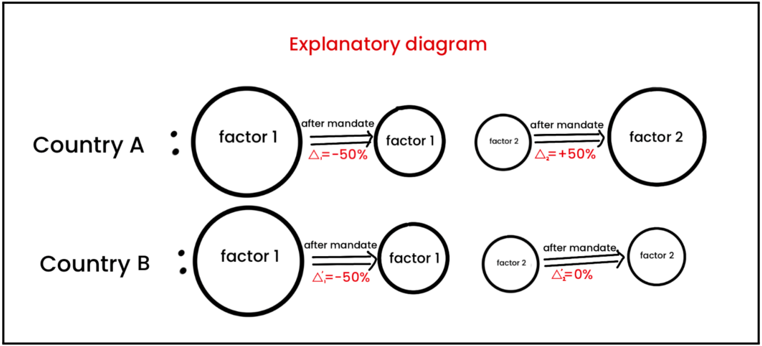
## 1. Project background

- During **COVID-19**, protective behaviours such as **mask wearing**, social distancing, hand hygiene, and self-isolation were important for reducing transmission.
- Compliance was shaped not only by policy rules, but also by risk perception, trust, demographic characteristics, wellbeing, and broader social context.
- These predictors may operate **differently across countries** because policy environments, institutions, culture, and public expectations vary.

## 2. Research questions

- Within each country, do the key predictors of compliance with COVID-19 protective behaviours change before and after mask mandates?
- Are these changes consistent across countries, or do they vary?
- Is there a set of common cross-national predictors, alongside country-specific predictors?

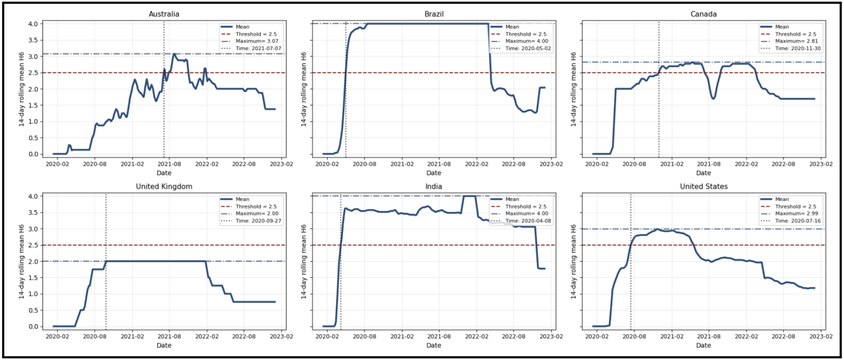


# Data Sources

## 1. Behaviour and policy datasets

- **YouGov dataset:** This dataset contains information relating to individuals' health protection behaviours, amongst other things
- **OxCGRT dataset:** This dataset contains the strictness with which the government has enforced its mask-wearing policy.
- Survey responses are linked to policy data by **country, region and date**.

## 2. Policy timing figure




This figure show how mask policy intensity changed over time, and how the mandate threshold was used to define the before and after mandate periods.

# Data Sources

## 3. Country coverage

The current cleaned cross-national sample includes six countries:

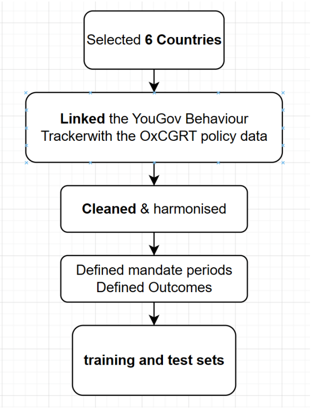
| Country code | Country name |
|---|---|
| AUS | Australia |
| BRA | Brazil |
| CAN | Canada |
| ~~CHN~~ | ~~China~~ |
| GBR | United Kingdom |
| IND | India |
| USA | United States |

China was excluded from the current analysis because several key variables needed for the harmonised workflow were not available.

# What Have I Done So Far?

## 1. Data pre-processing

- For the YouGov and OxCGRT files, various formatting issues have been resolved.
- Mapped survey regions to policy regions.
- Defined before/after mandate periods.
- Defined two outcomes: Face-mask-wearing and Overall-protective-behaviour.
- Cleaned the data: Removing columns and samples with too many missing values, encoding the data, making column names more readable, and splits.



# What Have I Done So Far?

## 2. About Model

### (1) Random Forest

- An ensemble of many decision trees trained on bootstrapped samples.
- Combines tree predictions to improve stability and reduce overfitting.

### (2) XGBoost

- A gradient-boosted tree model that builds trees sequentially.
- Each new tree improves the errors left by previous trees.

## Why these two models?

- In Ryan's study, these two tree-based models were the strongest performers.
- They capture nonlinear relationships and interactions among mixed survey predictors, while providing feature-importance outputs for cross-country comparison.

## 1. Cleaned sample size

The preprocessing pipeline produced cleaned before- and after-mandate datasets for all six countries.

| Country | Before mandate | After mandate | Total |
|---|---:|---:|---:|
| Australia | 15,757 | 25,988 | 41,745 |
| Brazil | 1,112 | 9,006 | 10,118 |
| Canada | 15,944 | 23,967 | 39,911 |
| United Kingdom | 7,838 | 24,580 | 32,418 |
| India | 1,736 | 13,629 | 15,365 |
| United States | 1,787 | 15,428 | 17,215 |

This shows that the pipeline has produced analysable data for each country and for both policy periods.

## 2. Train-test split

The next step was to prepare data for modelling.

- I split the data separately for each country, policy period, and outcome.
- I used an 80/20 train-test split.
- Stratification was used where possible to preserve the class balance of the outcome.

| Dimension | Values | Number |
|---|---|---|
| Countries | AUS, BRA, CAN, GBR, IND, USA | 6 |
| Policy periods | Before mandate, after mandate | 2 |
| Outcomes | Face mask, overall protective behaviour | 2 |
| Total modelling tasks | $6 \times 2 \times 2$ | 24 |

## 3. Model Performance

# Summary

## Summary of current progres

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## Next Step

- xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

## All data and code

All code can be found in my GitHub: https://github.com/ZixaunZhao/COVID-19-Protective-Behavior-Cross-National-Analysis.git